Name _____ _____ Baggett/Fall2026/207/Quiz1

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) The average rate of change of f on $[a, b]$ is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

True **False**

Why? _____ /3

2. (**3** pts) If $\lim_{x\to 2} f(x) = 5$, then $f(2) = 5$.

True **False**

Why? _____ /3

3. (**4** pts) Let $f(x) = 3 - x^2$. Form the difference quotient at $a = 2$ and simplify it as far as it will go:

$$\frac{f(2+h) - f(2)}{h}$$

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4. (**4** pts) A drone rises straight up. Its height above the ground after t seconds is

$$s(t) \;=\; 40t - 5t^2 \text{ meters}, \qquad 0 \le t \le 8.$$

(a) Find the drone's **average velocity** over the interval $[1, 3]$. Include units.

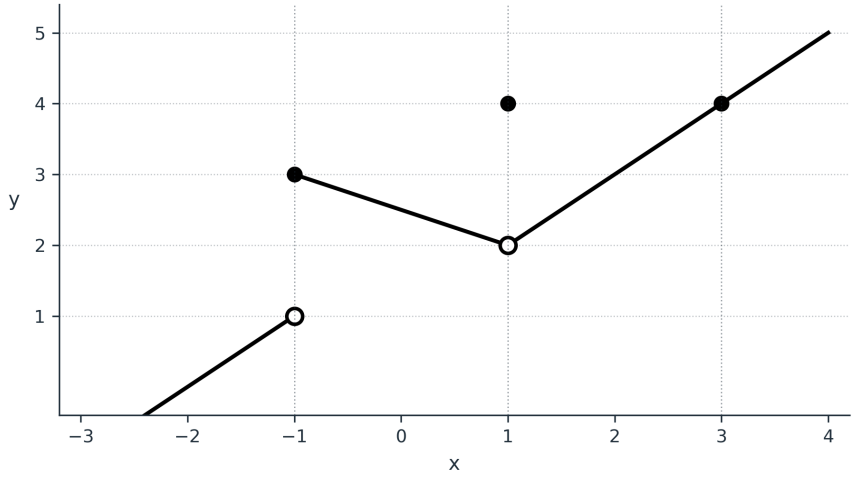
(b) In **one sentence**, say what that number tells you about the drone.

(c) Over $[1, 1.5]$, $[1, 1.1]$ and $[1, 1.01]$ the average velocities are 27.5, 29.5 and 29.95 m/sec. What number are they heading for, and what does it say about the drone at the **instant** $t = 1$?

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5. (**6** pts) Use the graph of $y = f(x)$ below. If a limit does not exist, say so.



(a) $\lim_{x \to -1^-} f(x) =$ _____ (b) $\lim_{x \to -1} f(x) =$ _____

(c) $\lim_{x \to 1} f(x) =$ _____ (d) $f(1) =$ _____

(e) $\lim_{x \to 3} f(x) =$ _____ (f) $f(3) =$ _____

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